YOLO vs FCOS Comparison Report

Generated: 2026-03-09

1. Metrics comparison

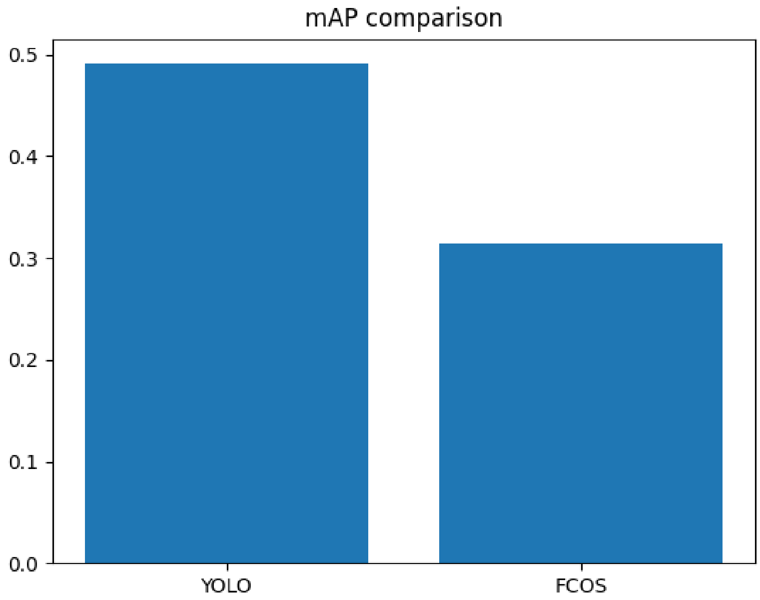
YOLO:
mAP = 0.4909388296425004
mAP50 = 0.7489650566051065
FPS = 8.086550404513403

FCOS:
mAP = 0.314
mAP50 = 0.441
FPS = 9.323673565851912

YOLO vs FCOS Comparison Report

Generated: 2026-03-09

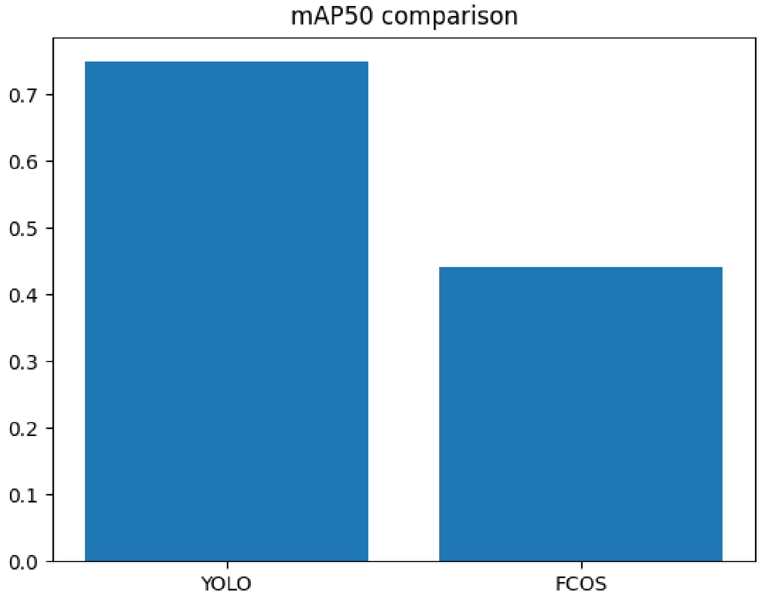
2. mAP comparison



YOLO vs FCOS Comparison Report

Generated: 2026-03-09

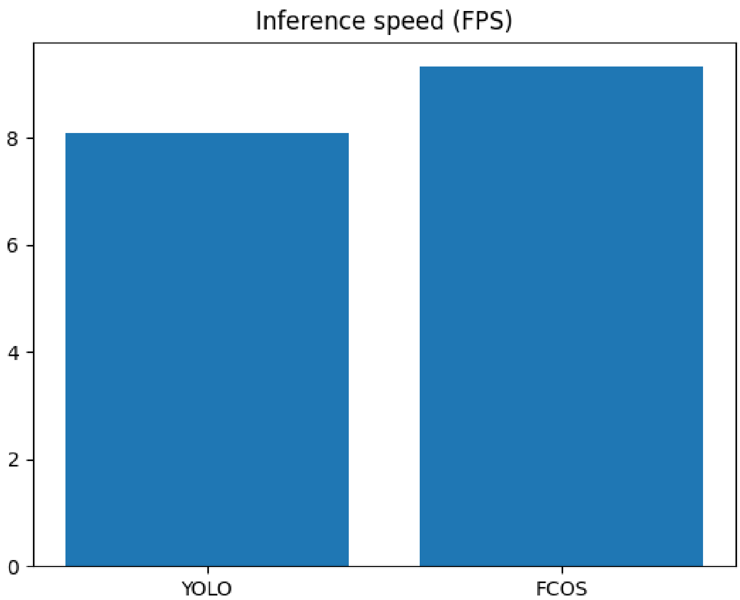
3. mAP50 comparison



YOLO vs FCOS Comparison Report

Generated: 2026-03-09

4. FPS comparison



YOLO vs FCOS Comparison Report

Generated: 2026-03-09

5. Conclusions

For FCOS:
The average precision across all objects is 31.4%, with mAP50 at 44.1%.
Inference speed is approximately 9 frames per second.

For a simple model trained for 12 epochs, the results can be considered satisfactory.
mAP for small objects is only 0.042 — very low, meaning small objects are almost not detected.
mAP for large objects is 0.44 — already quite decent.

For YOLO:

mAP	mAP50	FPS
0.490939	0.748965	8.086550

— YOLO achieved higher accuracy after 12 epochs.
This is unexpected, since on Windows with CPU for 1 epoch, YOLO showed much lower accuracy at a much higher speed.
On CUDA, even FCOS has higher speed — but with worse accuracy.
Overall, YOLO's accuracy at mAP 0.5 and mAP50 0.75 indicates fairly good detection with a soft threshold of 0.5.

Considering the above, if time permits, the model could be retrained to better detect small objects and account for class imbalance.